



**BITS Pilani**  
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Computer Vision  
Computer Vision (S2-25\_AIMLZG525)

## Session #2: Digital Image Fundamentals Raghavendra Singh

Acknowledgement: Slide Materials adopted from - Intro to Computer Vision (Cornell Tech); Noah Snavely

*S. P. Vimal* | *CSIS Department* | *WILP* | *BITS – Pilani* ([vimalsp@wilp.bits-pilani.ac.in](mailto:vimalsp@wilp.bits-pilani.ac.in))



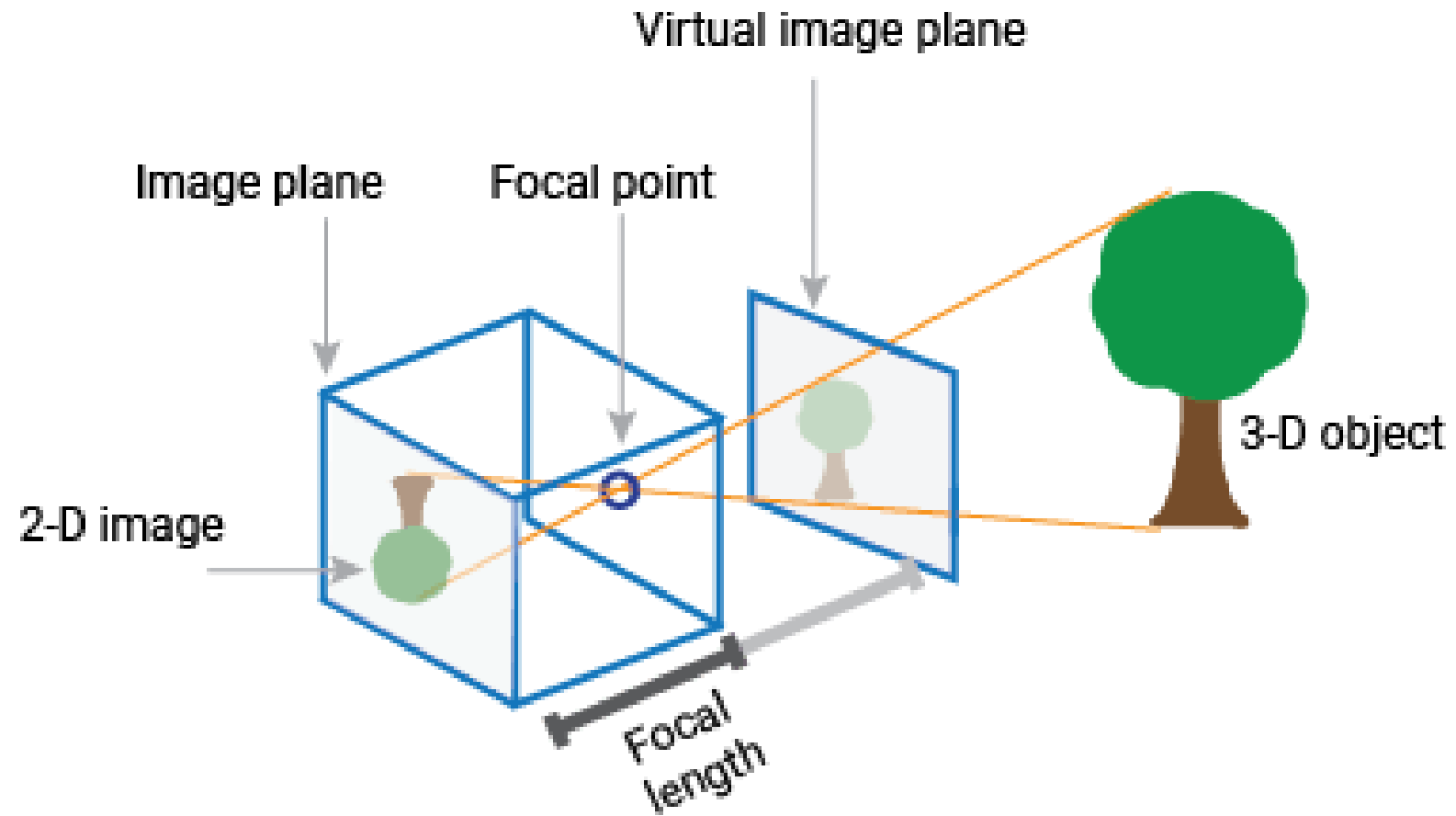
## Topics

- Image Representation
- Image Digitization ( Sampling & Quantization)
- Digital Image Properties
  
- OpenCV/Python Demonstration

Image representation and image analysis tasks, Image digitization - Sampling and resolution, Digital Images, Digital Image types -Binary, Gray-scale and Color, Color Images, Color spaces: RGB and HSV



# Simple Image Formation Model





**Pinhole Camera Model**, which is the fundamental geometric model used in computer vision and photography to describe how a 3D scene is projected onto a 2D surface.

### 1. The Geometry of Projection

- **3-D Object:** Represents the real world. Light rays travel from the object toward a single point of convergence.
- **Focal Point (Center of Projection):** This is the "pinhole." In a physical camera, all light rays must pass through this infinitesimal point.
- **Focal Length:** The distance between the Focal Point and the Image Plane. This distance determines how large the object will appear on the sensor and the field of view.

### 3. Key Takeaways

- **Dimension Reduction:** The model explains the mathematical transformation of coordinates from 3D  $(x, y, z)$  to 2D  $(u, v)$ .
- **Perspective:** Objects further away from the focal point appear smaller on the image plane because the angle of the light rays becomes narrower.

### 2. The Two Image Planes

The diagram shows two different ways to mathematically represent where the image "lives":

- **Image Plane (The "Real" Version):** Located *behind* the focal point. Notice that the tree here is **inverted** (upside down and flipped). This is how light actually behaves in a dark box or a real camera sensor.
- **Virtual Image Plane (The "Mathematical" Version):** Often placed in front of the focal point at the same focal distance. This is used in computer graphics and vision because the image is **upright**, making the math easier to visualize and calculate while maintaining the same perspective geometry.



# Mathematical equations (Perspective Projection) used to calculate the 2D coordinates from the 3D position

To derive the 2D coordinates from a 3D point, we use **Similar Triangles**.

Based on the diagram you provided, we can map a point in 3D space  $(X, Y, Z)$  to a point on the 2D image plane  $(x, y)$ .

## 1. The Geometry of Similar Triangles

Imagine looking at the setup from the side (the  $Y - Z$  plane). The ray of light forms two similar triangles: one between the focal point and the object, and another between the focal point and the image plane.

Using the properties of similar triangles, the ratio of the height to the distance from the focal point must be constant:

$$\frac{x}{f} = \frac{X}{Z} \quad \text{and} \quad \frac{y}{f} = \frac{Y}{Z}$$

## 2. The Projection Equations

By rearranging the terms above, we get the standard equations for perspective projection. This calculates where a 3D point will land on your 2D sensor:

- **Horizontal Coordinate:**  $x = f \cdot \frac{X}{Z}$
- **Vertical Coordinate:**  $y = f \cdot \frac{Y}{Z}$

**Note:** If you are using the **Real Image Plane** (behind the focal point), the coordinates would technically be negative ( $x = -f \frac{X}{Z}$ ), representing the inverted image. For the **Virtual Image Plane** shown in your slide, the coordinates remain positive.

## 3. Matrix Form (Homogeneous Coordinates)

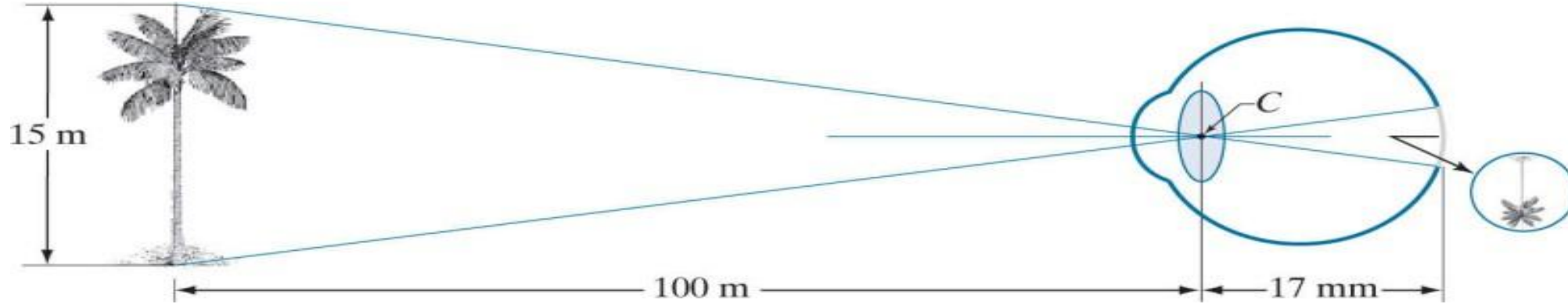
In computer vision, we typically express this transformation using a **Projection Matrix**. This allows us to handle the division by  $Z$  linearly using homogeneous coordinates:

$$\begin{bmatrix} x_h \\ y_h \\ w \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

To get the final 2D pixel coordinates, you simply divide the result by the scale factor  $w$  (which, in this basic case, is equal to  $Z$ ).



# Image Formation in the Eye



This slide applies the **Pinhole Camera Model** to the human eye, demonstrating how physical dimensions in the real world translate to an image size on the retina.

## Key Components of the Diagram

- **The Object:** A palm tree with a height of 15 m.
- **Object Distance ( $Z$ ):** The tree is located 100 m away from the eye's optical center ( $C$ ).
- **Focal Length ( $f$ ):** The distance from the optical center to the retina is given as 17 mm.
- **Image Formation:** Light rays cross at the center ( $C$ ), resulting in an **inverted image** on the back of the eye (retina).

## The Mathematical Calculation

To find the height of the image on the retina ( $h$ ), we use the similar triangles formula discussed previously:

$$\frac{\text{Image Height}}{\text{Focal Length}} = \frac{\text{Object Height}}{\text{Object Distance}}$$

Substituting the values from the slide:

$$\frac{h}{17 \text{ mm}} = \frac{15 \text{ m}}{100 \text{ m}}$$

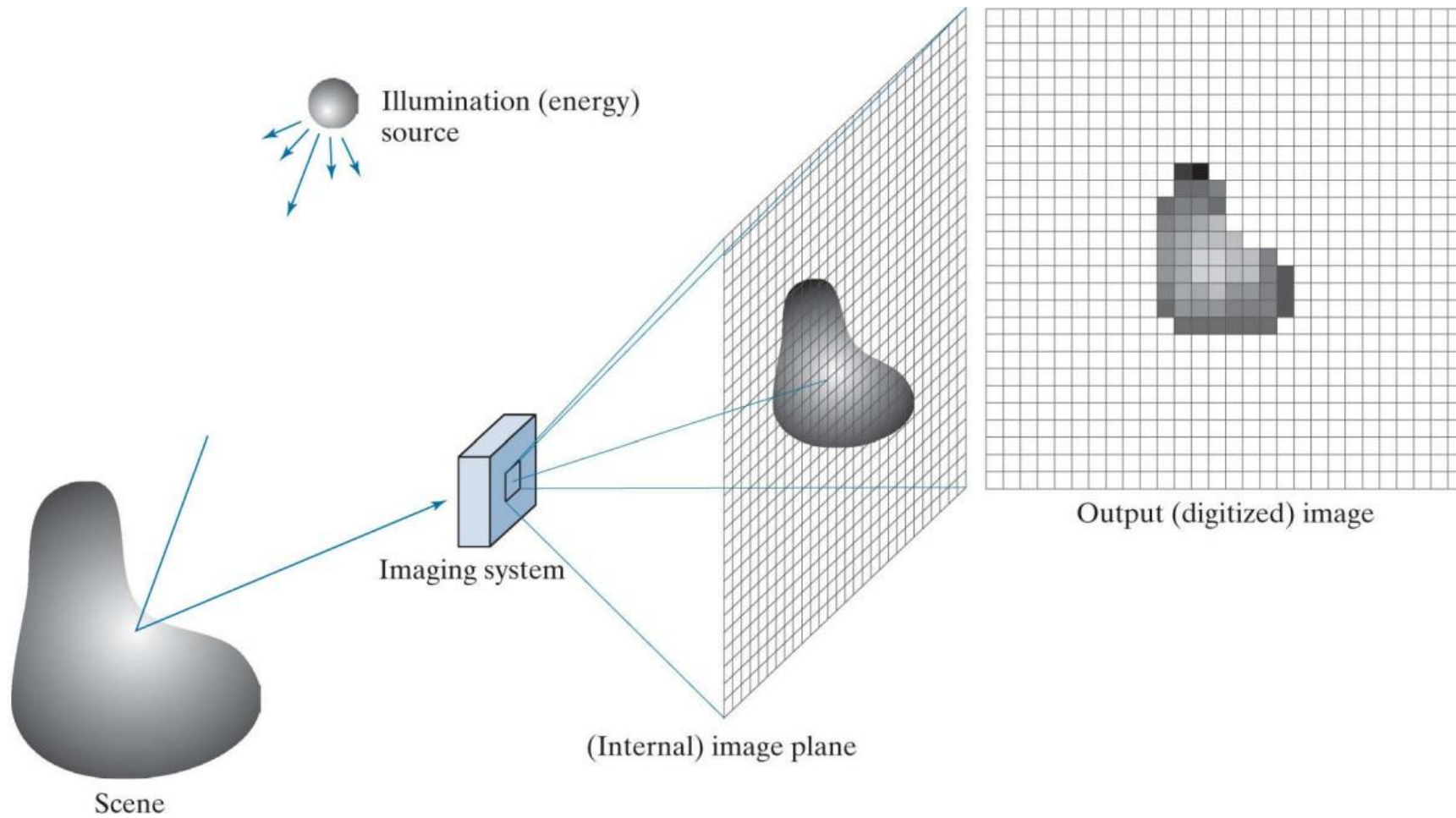
$$h = 17 \text{ mm} \cdot 0.15 = \mathbf{2.55 \text{ mm}}$$

## Summary

- The eye acts as a biological camera where the **lens center** is the focal point and the **retina** is the image plane.
- Despite the tree being massive (15 m), the geometry of perspective projection reduces it to a tiny 2.55 mm image on the retina.



# Digital Image Acquisition







# Digital Image Acquisition

This slide illustrates the fundamental process of **Digital Image Acquisition**, showing how a physical object in the real world is converted into a digital format that a computer can understand.

Think of it as the "life cycle" of a photo, moving from light to bits. Here is a breakdown of the four key stages shown:

## 1. Illumination and the Scene

The process starts at the bottom left. For us to "see" anything, there must be an **Illumination Source** (like the sun or a lightbulb).

- Energy from the source hits the **Scene** (the physical object).
- The object reflects a portion of that energy toward the camera. This reflected light carries the information about the object's shape and color.

## 2. The Imaging System

The light passes through an **Imaging System** (the camera lens and body).

- This system collects the incoming light and focuses it.
- It acts as a bridge, projecting the 3D scene onto a 2D surface.

## 4. Output (Digitized) Image

The final result is the **Digital Image**. At this stage, two things have occurred:

- **Sampling:** The grid has determined the spatial resolution (how many pixels make up the image).
- **Quantization:** Each pixel is assigned a numerical value representing the brightness/intensity of the light at that specific spot.

**Summary:** The "smooth" real world is chopped up into a grid of squares, where each square is assigned a number. That grid of numbers is what your computer stores as an image file.





# Simple Image Formation Model

- We denote images by two-dimensional functions of the form  $f(x, y)$ .
  - Value of  $f$  at spatial coordinates  $(x, y)$  is a scalar quantity;
  - Physical meaning of  $f(x, y)$  is determined by the source of the image, and whose values are proportional to energy radiated by a physical source (e.g., electromagnetic waves);

$$0 \leq f(x, y) < \infty$$

- Function  $f(x, y)$  is characterized by two components:
  - Amount of source illumination incident on the scene being viewed, and
  - Amount of illumination reflected by the objects in the scene.

$$f(x, y) = i(x, y)r(x, y)$$

$$0 \leq i(x, y) < \infty$$

$$0 \leq r(x, y) \leq 1$$



## Simple Image Formation Model - Grayscale Image

- Let the intensity (gray level) of a monochrome image at any coordinates  $(x, y)$  be denoted by

$$\ell = f(x, y) \quad L_{\min} \leq \ell \leq L_{\max}$$

$$L_{\min} = i_{\min} r_{\min} \quad L_{\max} = i_{\max} r_{\max}$$

- For a gray scale, the range is scaled between  $[0, L-1]$  ;
  - All intermediate values from 0 to  $L-1$  are shades of gray varying from black to white.



# Explanation - Simple Image Formation Model - Grayscale Image

## Core Mathematical Representation

- **Intensity Function:** The intensity (gray level) of a monochrome image at any spatial coordinate  $(x, y)$  is denoted by the function  $\ell = f(x, y)$ .
- **Intensity Range:** The value of  $\ell$  is bounded by a minimum and maximum limit:  $L_{\min} \leq \ell \leq L_{\max}$ .

## The Components of Intensity

While not fully defined in text on this specific slide, the variables  $i$  and  $r$  refer to the two physical factors that create an image:

- **Illumination ( $i$ ):** The amount of source light incident on the scene.
- **Reflectance ( $r$ ):** The amount of light reflected by the objects in the scene.
- **The Relationship:** The total intensity at any point is the product of these two:  $L = i \cdot r$ . Therefore, the total range is determined by the product of their respective minimums and maximums ( $L_{\min} = i_{\min} r_{\min}$  and  $L_{\max} = i_{\max} r_{\max}$ ).

## Digital Scaling

To make this useful for computers, the continuous physical light levels are mapped to a discrete digital scale:

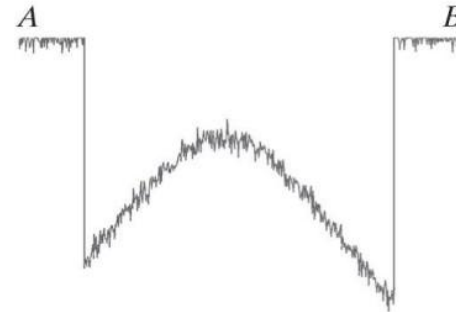
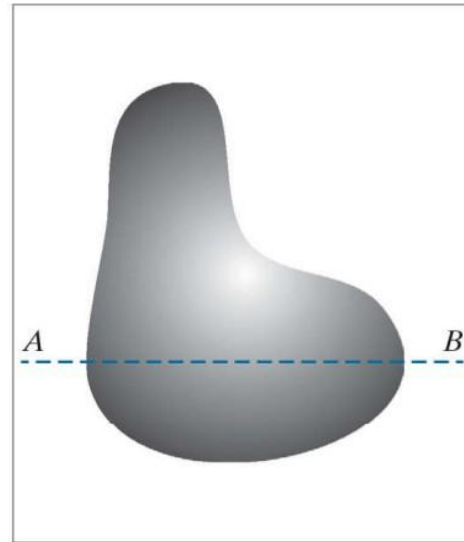
- **Scale Range:** The intensity range is scaled between  $[0, L]$ .
- **Shades of Gray:** In this digital scale, **0** typically represents total **black**, and **L-1** represents total **white**.
- **Intermediate Values:** All values between 0 and  $L - 1$  represent varying shades of gray.

**Common Example:** In a standard 8-bit image,  $L = 256$ . The scale runs from **0 (black)** to **255 (white)**, providing 256 distinct levels of gray.



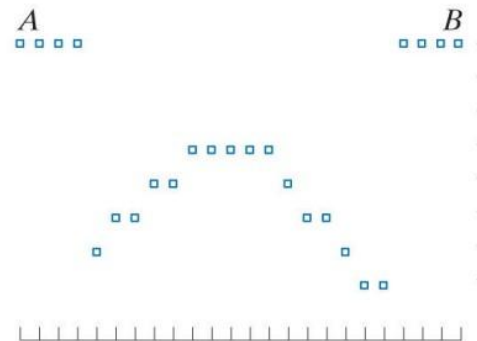
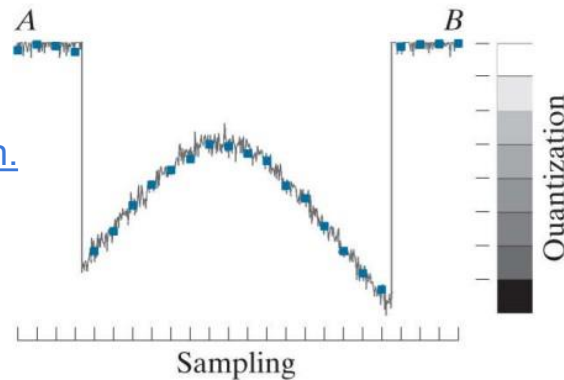
# Sampling and Quantization

Continuous image



A scan line showing intensity variations along line AB in the continuous image

Sampling and quantization.



Digital scan line

Digitizing the coordinate values is called *sampling* ; Digitizing the amplitude values is called *quantization*.



# Explanation - Sampling and Quantization

The fundamental process of converting a real-world, continuous image into a digital format that a computer can understand. This process is divided into two main steps: **Sampling** and **Quantization**.

## 2. The Digitization Process (Bottom Row)

To make this data digital, we have to "chunk" it in two ways:

- **Sampling (Horizontal axis):** This is the process of digitizing **coordinates**. Instead of looking at every infinite point along the line  $AB$ , we only take measurements at specific intervals (represented by the tick marks on the bottom axis labeled "Sampling").
  - *Analogy:* Like taking a snapshot every 1 meter rather than recording every millimeter.

## 3. The Digital Scan Line (Bottom Right)

The final result is a series of **discrete dots**. These dots represent the pixels in a digital image. Each dot has:

1. A specific **position** (defined by Sampling).
2. A specific **color/brightness value** (defined by Quantization).

- **Quantization (Vertical axis):** This is the process of digitizing **amplitude (brightness)**. Even after sampling a point, its brightness could be any value (e.g., 0.7342...). Quantization forces that value into a discrete level (e.g., "Level 7" on a grayscale).
  - *Analogy:* Like rounding a price to the nearest dollar.

## Key Takeaway

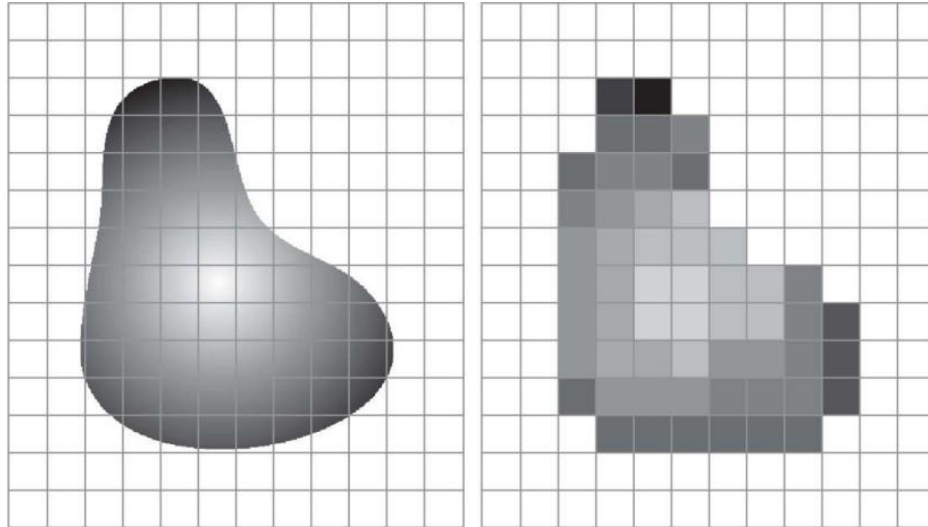
The slide summarizes the core definitions at the bottom:

- **Sampling:** Digitizing the **coordinate** values (Spatial resolution).
- **Quantization:** Digitizing the **amplitude** values (Gray-level resolution).



# Sampling and Quantization

Continuous image projected onto a sensor array



Result of image sampling and quantization

- **Sampling:** This is the process of digitizing **coordinate** values (also known as spatial resolution). It determines how many pixels or data points are taken across the image.
- **Quantization:** This is the process of digitizing **amplitude** values (also known as gray-level resolution). It determines the number of discrete intensity levels (colors or shades of gray) assigned to each sampled point.

Essentially, **sampling** decides the grid (where the pixels are), while **quantization** decides the value (what color/shade each pixel is).





# Matrix Representation of an Image

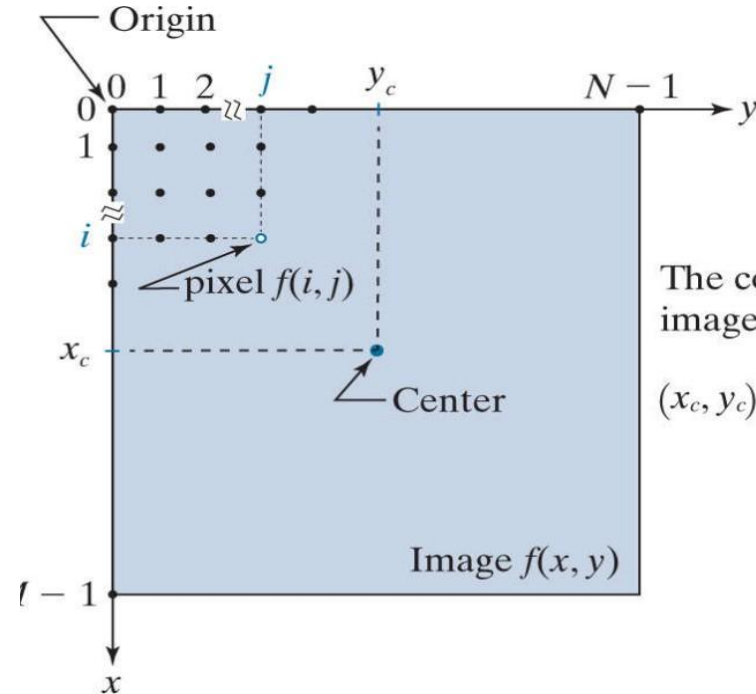
$$f(x, y) = \begin{bmatrix} f(0, 0) & f(0, 1) & \dots & f(0, N-1) \\ f(1, 0) & f(1, 1) & \dots & f(1, N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1, 0) & f(M-1, 1) & \dots & f(M-1, N-1) \end{bmatrix}$$

## 1. The Matrix Representation

On the left, you see the mathematical notation for an image of size  $M \times N$  (M rows and N columns).

- $f(x, y)$ : Represents the intensity or color value at a specific pixel.
- **Indices:** Notice that the indexing starts at 0.
  - Rows go from 0 to  $M - 1$ .
  - Columns go from 0 to  $N - 1$ .

- **The Origin (0, 0):** Located at the **top-left corner** of the image.
- **x-axis:** Points **downward** (representing rows).
- **y-axis:** Points to the **right** (representing columns).
- **Pixel  $f(i, j)$ :** This is the value located at the intersection of the  $i^{th}$  row and the  $j^{th}$  column.



The coordinates of the image center are

$$(x_c, y_c) = \left( \text{floor}\left(\frac{M}{2}\right), \text{floor}\left(\frac{N}{2}\right) \right)$$

$$(x_c, y_c) = \left( \text{floor}\left(\frac{M}{2}\right), \text{floor}\left(\frac{N}{2}\right) \right)$$

**Example:** If you have a  $100 \times 100$  image, the center pixel would be at (50, 50). If the image is  $5 \times 5$ , the center would be at (2, 2).

# Spatial & Intensity Resolution

## Two fundamental properties of digital images:

- **Spatial resolution** is a measure of the smallest discernible detail within an image. It essentially describes how many "dots" or pixels are used to represent a physical area.
- **Intensity resolution** is the smallest discernible change in the intensity level (the brightness or color shade) of a pixel.

930 dpi

High detail; the numbers and fine lines on the clock are sharp and clear.

300 dpi

Slightly less sharp, but still very clear for standard viewing.

150 dpi

The image starts to look "soft" or slightly blurry as details begin to merge.

72 dpi

Significant loss of detail; "pixelation" occurs, making small text or fine lines difficult to read.

930 dpi



300 dpi



150 dpi



72 dpi





# Spatial & Intensity Resolution

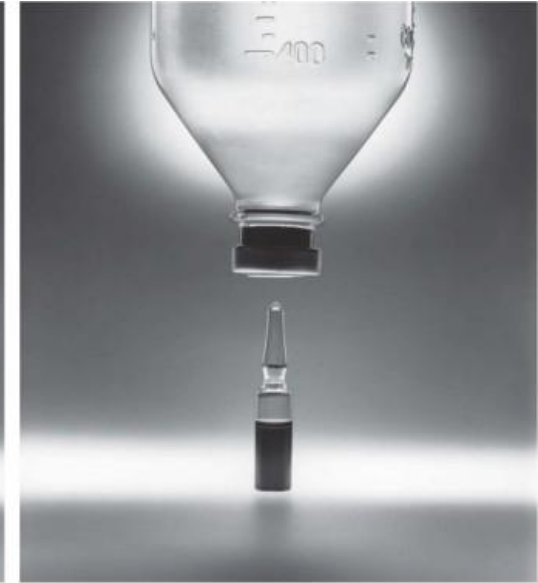
The second slide uses images of a laboratory bottle and ampoule to demonstrate **intensity levels**. While the spatial resolution remains constant ( $2022 \times 1800$  pixels), the number of possible brightness shades changes:

- **256-level:** This is typically an 8-bit image, offering smooth transitions and high detail in shadows and highlights.
- **32-level:** At this lower intensity resolution, you might begin to see **false contouring**—where smooth gradients (like the light on the wall) appear as distinct "bands" or blocks of gray because there aren't enough shades to represent the subtle change.
- **128 and 64-level:** These represent the middle ground where the reduction in quality is present but less jarring than the 32-level version.

256-level



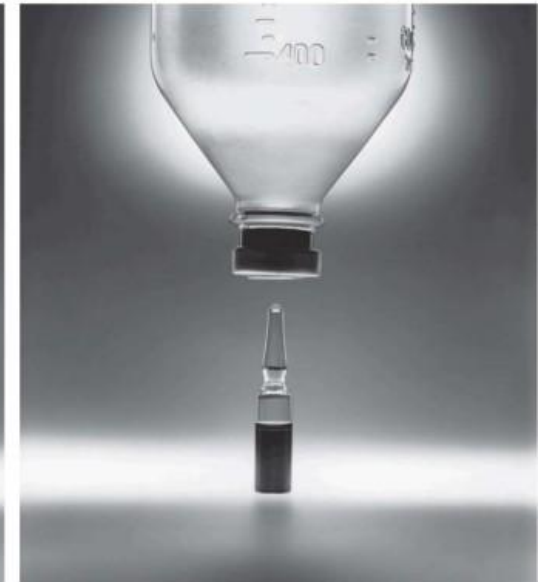
128-level



64-level



32-level



All images have spatial resolution  $2022 \times 1800$



# Spatial & Intensity Resolution

## 32 to 16-level

The image begins to show **false contouring**, where smooth transitions (like the light on the wall) appear as distinct bands.

## 8 to 4-level

Major loss of detail; the image looks "posterized" as the camera is forced to group many different shades into a few blocks.

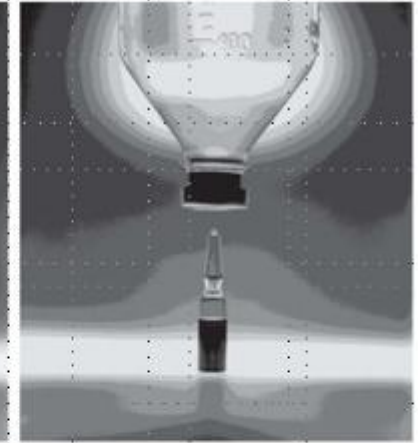
## 2-level

The image becomes "binary" (black and white only), losing almost all subtle texture and depth.

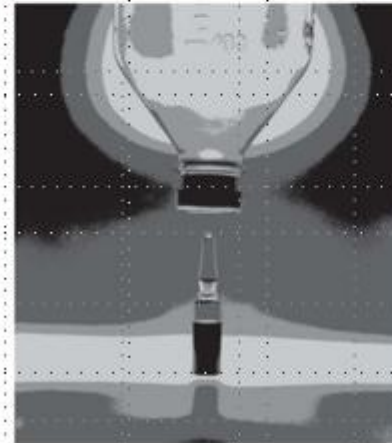
16-level



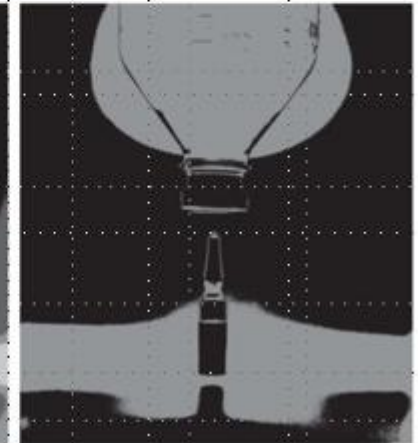
8-level



4-level



2-level







- Readings:

Digital Image Processing, Rafael C. Gonzalez & Richard E woods, Third Ed, Chapter 2